Nomvelo Zulu

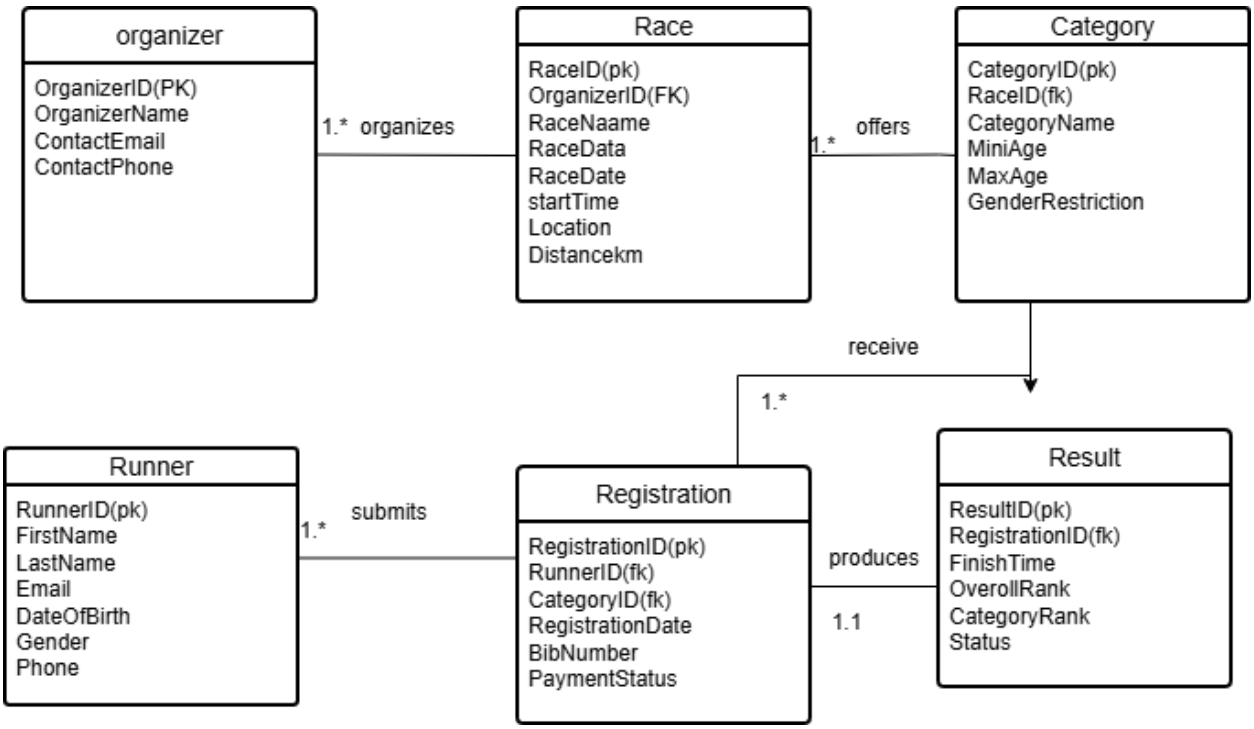
St10469528

Programming6212(POE Part 1)

Due Date : 04 September 2026

Section A

Race Day System-Entity Relationship Diagram



SECTION B

HTTP Method	Route	Description	Role Required	Request	Expected Response
Get	/api/races	Lists all races on the system.	None	None	200 OK — array of race objects
Get	/api/races/{id}	Returns full details for one race.	None	None	200 OK — race object; 404 Not Found — race doesn't exist
Post	/api/races	Creates a new race, linked to an organizer.	Admin	{ raceName, raceDate, startTime, location, distanceKm, organizerId }	201 Created — new race record; 400 Bad Request — invalid/missing fields
Put	/api/races/{id}	Updates an existing race's details.	Admin	{ raceName, raceDate, startTime, location, distanceKm }	200 OK — updated race object; 404 Not Found
Delete	/api/races/{id}	Removes a race and its dependent categories.	Admin	None	204 No Content; 404 Not Found
Get	/api/races/{id}/categories	Lists the categories offered for a race (e.g. 5K, 10K).	None	None	200 OK — array of categories

Post	/api/races/{id}/categories	Adds a new category to a race.	Admin	{ categoryName, minAge, maxAge, genderRestriction }	201 Created — new category record; 404 Not Found — race doesn't exist
Post	/api/runners	Registers a new runner account.	None (public)	{ firstName, lastName, email, dob, gender, phone }	201 Created — new runner record; 409 Conflict — email already registered
Get	/api/runners/{id}	Returns a runner's profile.	Runner (own) or Admin	None	200 OK — runner object; 403 Forbidden; 404 Not Found
Post	/api/races/{id}/register	Registers a logged-in runner into a race category and issues a bib number.	Runner	{ runnerId, categoryId }	201 Created — registration record with bib number; 404 Not Found — race/category doesn't exist; 409 Conflict — already registered

Get	/api/runners/{id}/registrations	Lists all race registrations for a runner.	Runner (own) or Admin	None	200 OK — array of registrations
Delete	/api/registrations/{id}	Cancels an existing registration.Cancels an existing registration.	Runner (own) or Admin	None	204 No Content; 404 Not Found
Post	/api/registrations/{id}/result /api/races/{id}/results	Records a runner's finish result for their registration.	Organizer/Admin	{ finishTime, overallRank, categoryRank, status }	201 Created — result record; 404 Not Found — registration doesn't exist; 409 Conflict — result already submitted
Get	/api/races/{id}/results	Returns the leaderboard/results for a race, sorted by rank.	None	None	200 OK — array of result objects

SECTION C

```
Create Database RaceDayDB;
use RaceDayDB;

Create table Organizer(
OrganizerID int identity(1,1) primary key,
OrganizerName VARCHAR(150) NOT NULL,
ContactEmail VARCHAR(90) NOT NULL,
ContactPhone VARCHAR(10) NOT NULL
);

Create table Race(
RaceID int identity(1,1) primary key,
OrganizerID int,
RaceName VARCHAR(100) NOT NULL,
```

```
RaceDate DATE NOT NULL,
StartTime TIME NOT NULL,
Location VARCHAR(250) NOT NULL,
DistanceKm DECIMAL(5,2) NOT NULL,
FOREIGN KEY (OrganizerID) REFERENCES Organizer(OrganizerID)
);

Create table Category(
CategoryID int identity(1,1) primary key,
RaceID int,
CategoryName VARCHAR(100) NOT NULL,
MinAge INT NOT NULL,
MaxAge INT NOT NULL,
GenderRestriction VARCHAR(20) NOT NULL,
FOREIGN KEY(RaceID) REFERENCES Race(RaceID)
);

Create table Runner(
RunnerID int identity(1,1) primary key,
FirstName VARCHAR(70) NOT NULL,
LastName VARCHAR(70) NOT NULL,
Email VARCHAR(90) UNIQUE NOT NULL,
DateOfBirth DATE NOT NULL,
Gender VARCHAR(20) NOT NULL,
Phone VARCHAR(10) NOT NULL
);
```

```
Create table Registration(
RegistrationID int identity(1,1) primary key,
RunnerID int,
CategoryID int,
RegistrationDate DATE NOT NULL,
BibNumber VARCHAR(10) NOT NULL,
PaymentStatus VARCHAR(20) NOT NULL CHECK (PaymentStatus IN ('Paid', 'Pending', 'Refunded')),
FOREIGN KEY(RunnerID) REFERENCES Runner(RunnerID),
FOREIGN KEY(CategoryID) REFERENCES Category(CategoryID)
);

Create table Result(
ResultID int identity(1,1) primary key,
RegistrationID int UNIQUE,
FinishTime VARCHAR(10),
OverallRank INT,
CategoryRank INT,
Status VARCHAR(20) NOT NULL,
FOREIGN KEY(RegistrationID) REFERENCES Registration(RegistrationID)
);

-- POPULATE TABLES

-- Organizer
```

```
INSERT INTO Organizer (OrganizerName, ContactEmail, ContactPhone) VALUES
('Cape Town Athletics Club', 'info@ctac.co.za', '0211234567'),
('Johannesburg Runners Assoc', 'contact@jra.co.za', '0117654321'),
('Durban Coastal Runners', 'hello@dcr.co.za', '0319988776');

-- Race
INSERT INTO Race (OrganizerID, RaceName, RaceDate, StartTime, Location, DistanceKm) VALUES
(1, 'Cape Town Peace Run', '2026-10-10', '06:00:00', 'Green Point, Cape Town', 10.00),
(1, 'Table Mountain Trail Run', '2026-11-15', '05:30:00', 'Table Mountain, Cape Town', 21.10),
(2, 'Joburg City Marathon', '2026-09-20', '06:00:00', 'Sandton, Johannesburg', 42.20),
(3, 'Durban Beachfront Fun Run', '2026-10-05', '07:00:00', 'North Beach, Durban', 5.00);

-- Category
INSERT INTO Category (RaceID, CategoryName, MinAge, MaxAge, GenderRestriction) VALUES
(1, 'Open 10km', 18, 65, 'Mixed'),
(1, 'Junior 10km', 12, 17, 'Mixed'),
(2, 'Half Marathon Open', 18, 60, 'Mixed'),
(3, 'Full Marathon Open', 18, 60, 'Mixed'),
(3, 'Full Marathon Veteran', 45, 70, 'Mixed'),
(4, 'Fun Run Open', 10, 70, 'Mixed');

-- Runner
INSERT INTO Runner (FirstName, LastName, Email, DateOfBirth, Gender, Phone) VALUES
('Thabo', 'Mokoena', 'thabo.mokoena@email.com', '1995-03-14', 'Male', '0821234567'),
```

('Sarah', 'Naidoo', 'sarah.naidoo@email.com', '1998-07-22', 'Female', '0839876543'),

('Johan', 'van der Merwe', 'johan.vdm@email.com', '1988-11-02', 'Male', '0824567890'),

('Lerato', 'Dlamini', 'lerato.dlamini@email.com', '2001-05-19', 'Female', '0837654321'),

('Michael', 'Botha', 'michael.botha@email.com', '1975-01-30', 'Male', '0825551234'),

('Nomvula', 'Khumalo', 'nomvula.khumalo@email.com', '1993-09-08', 'Female', '0836667788');

-- Registration

INSERT INTO Registration (RunnerID, CategoryID, RegistrationDate, BibNumber, PaymentStatus) VALUES

(1, 1, '2026-09-01', '1001', 'Paid'),

(2, 1, '2026-09-02', '1002', 'Paid'),

(3, 3, '2026-08-20', '2001', 'Paid'),

(4, 4, '2026-08-25', '3001', 'Pending'),

(5, 5, '2026-08-26', '3002', 'Paid'),

(6, 6, '2026-09-03', '4001', 'Refunded');

-- Result

INSERT INTO Result (RegistrationID, FinishTime, OverallRank, CategoryRank, Status) VALUES

(1, '00:45:12', 3, 1, 'Finished'),

(2, '00:48:30', 7, 2, 'Finished'),

(3, '01:35:47', 1, 1, 'Finished'),

(4, '04:15:22', 15, 5, 'Finished'),

(5, '04:02:10', 10, 1, 'Finished'),

(6, NULL, NULL, NULL, 'DNS');

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